DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
Year 6

HIGHER 2 GEOGRAPHY **9173/02**

INSERT

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

Land use map of survey area along Sungei Bedok



Legend

Colour	Zoning
	Business
	Civic and Community Institution
	Commercial & Residential
	Education Institution
	Park
	Residential
	Transport Facilities
	Waterbody

Scale
1cm : 100m

Resource 2 for Question 1

Photograph of section of Sungei Bedok taken on 15 August



Resource 3 for Question 1

Bi-polar survey crafted by the students

Location:				**Date/Time:**		
Low Risk	**+2**	**+1**	**0**	**-1**	**-2**	**High Risk**
High percentage of vegetation cover						*Low percentage of vegetation cover*
High percentage of permeable surfaces						*Low percentage of permeable surfaces*
Lowest occupied floor of building is above channel height at bankfull discharge						*Lowest occupied floor of building is below channel height at bankfull discharge*
Presence of drainage system						*Absence of drainage system*
Total score:						

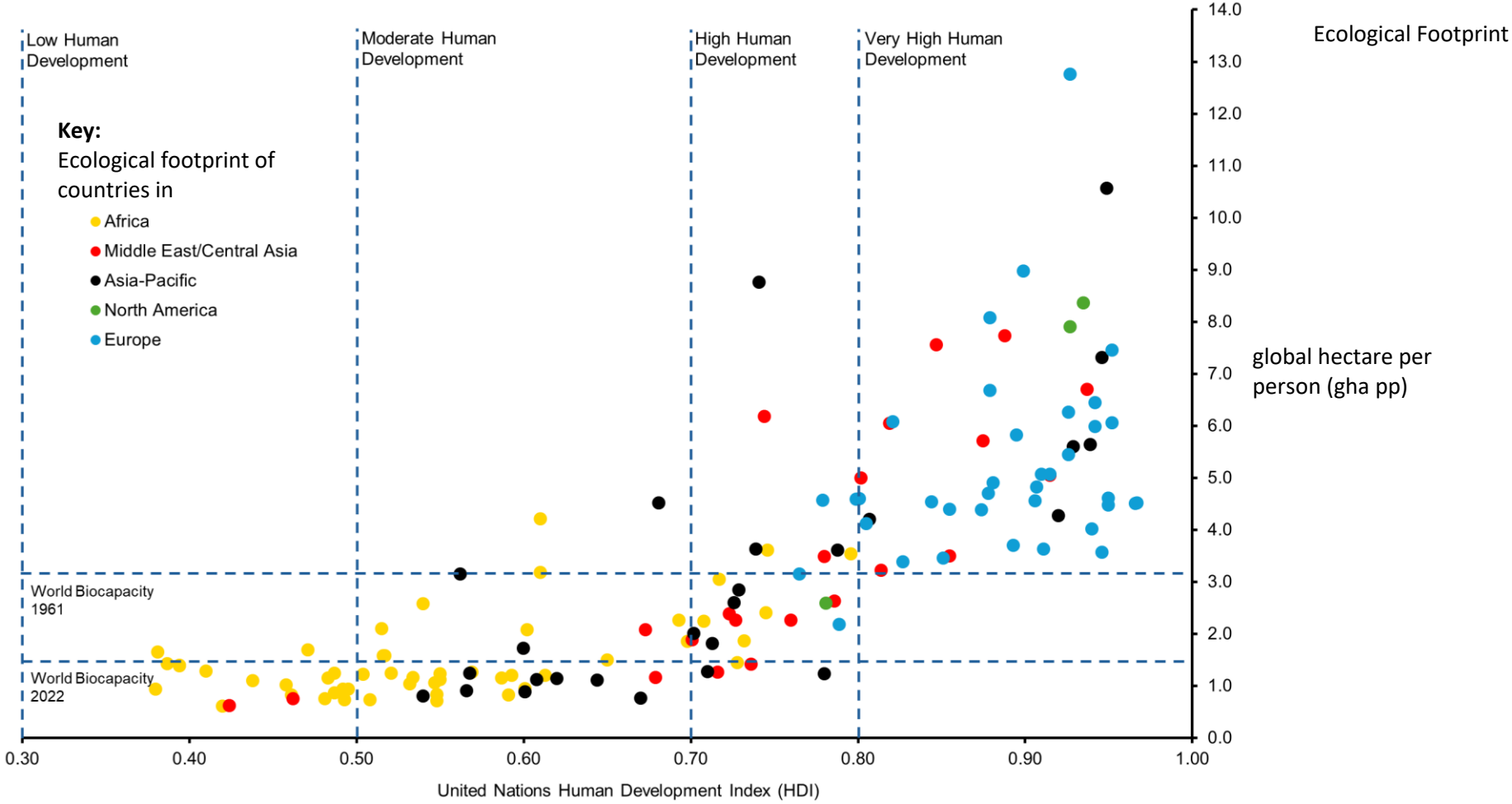
Resource 4 for Question 1

Bi-polar survey results for all land uses

Land use	**Vegetation**	**Permeable surfaces**	**Floor level vs bankfull**	**Drainage**	**Net Score**
Business	-2	-2	0	+1	**-3**
Civic & Community Institution	+1	+1	+1	-1	**+2**
Commercial & Residential	-1	-1	0	0	**-2**
Education Institution	+1	+1	+1	0	**+3**
Park	+2	+2	0	0	**+4**
Residential	0	-1	+1	-1	**-1**
Transport Facilities	-2	-2	0	0	**-4**

Resource 5 for Question 2

World Biocapacity* in 1961 and 2022, Ecological Footprint and Human Development Index (HDI) of countries in 2022**

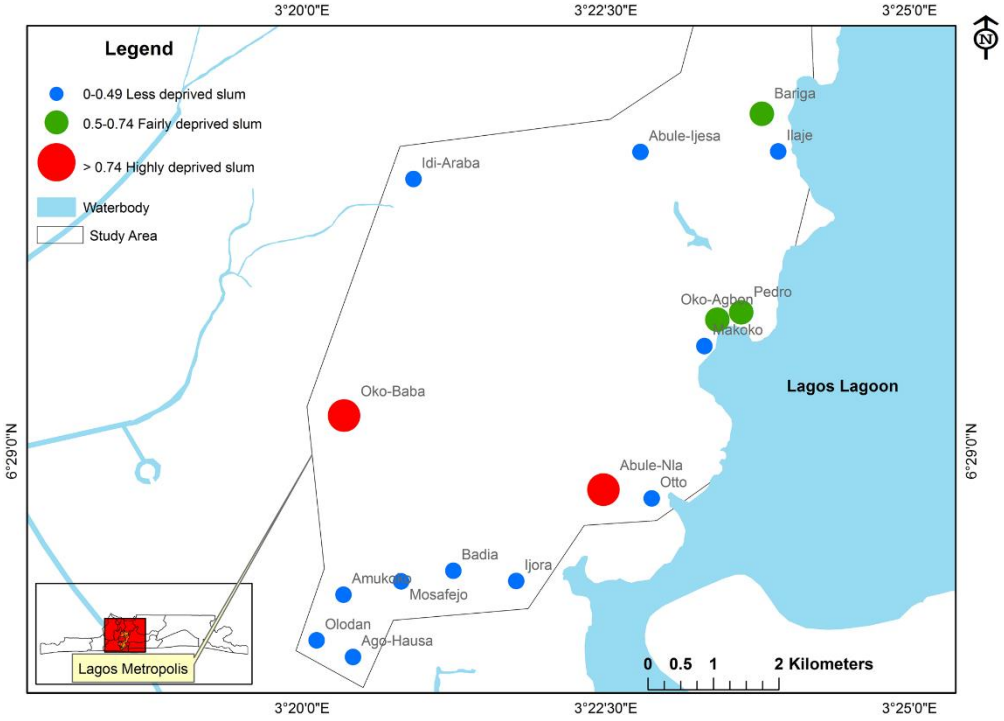


**Biocapacity is the capacity of biologically productive areas to provide for human demand. A value below the biocapacity is feasible for environmental regeneration.*

***HDI is a measure of achievement of development in the areas of life expectancy, literacy and economic growth. A high value indicates greater achievement.*

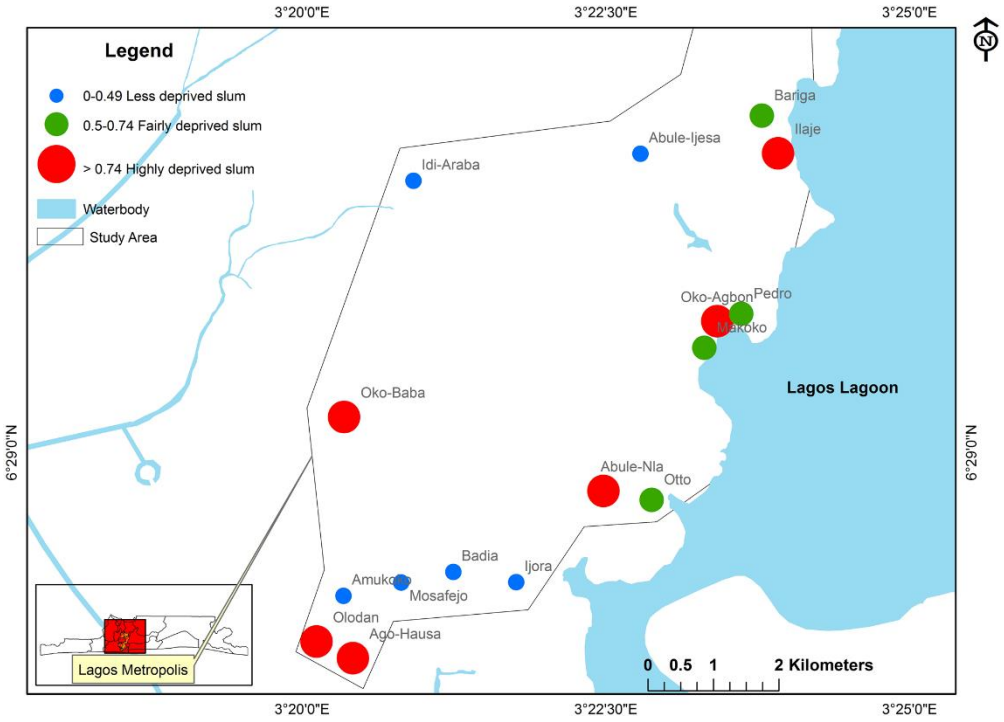
Resource 6A for Question 2

Performance of sanitation indicator of the slum deprivation index Lagos metropolis, Nigeria



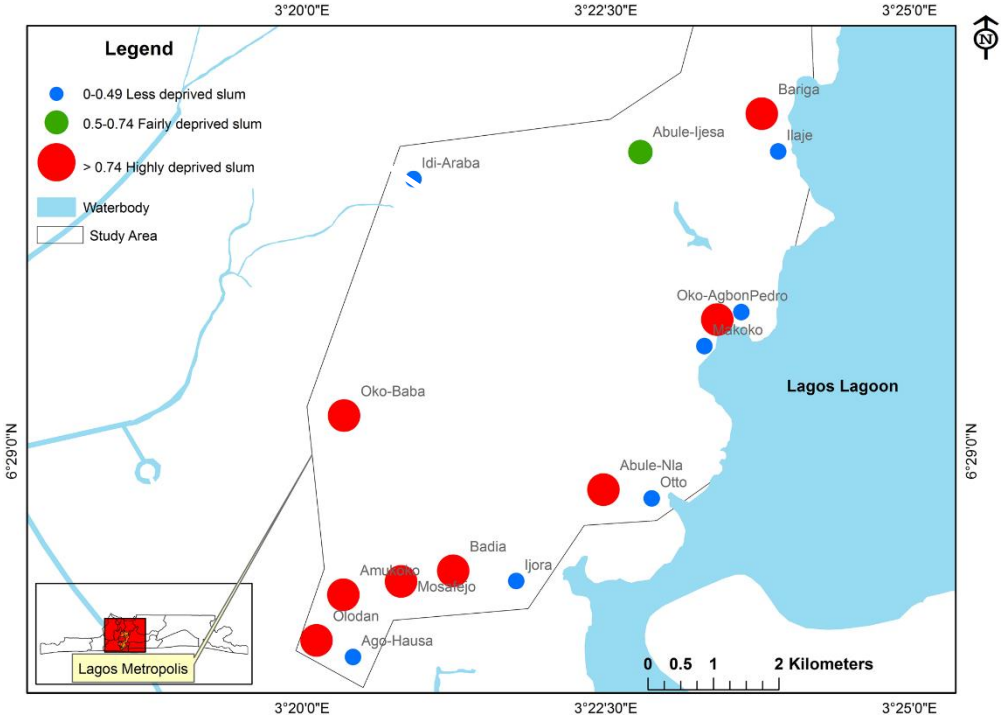
Resource 6B for Question 2

Performance of waste disposal indicator of the slum deprivation index Lagos metropolis, Nigeria



Resource 6C for Question 2

Performance of dwellings indicator of the slum deprivation index Lagos metropolis, Nigeria



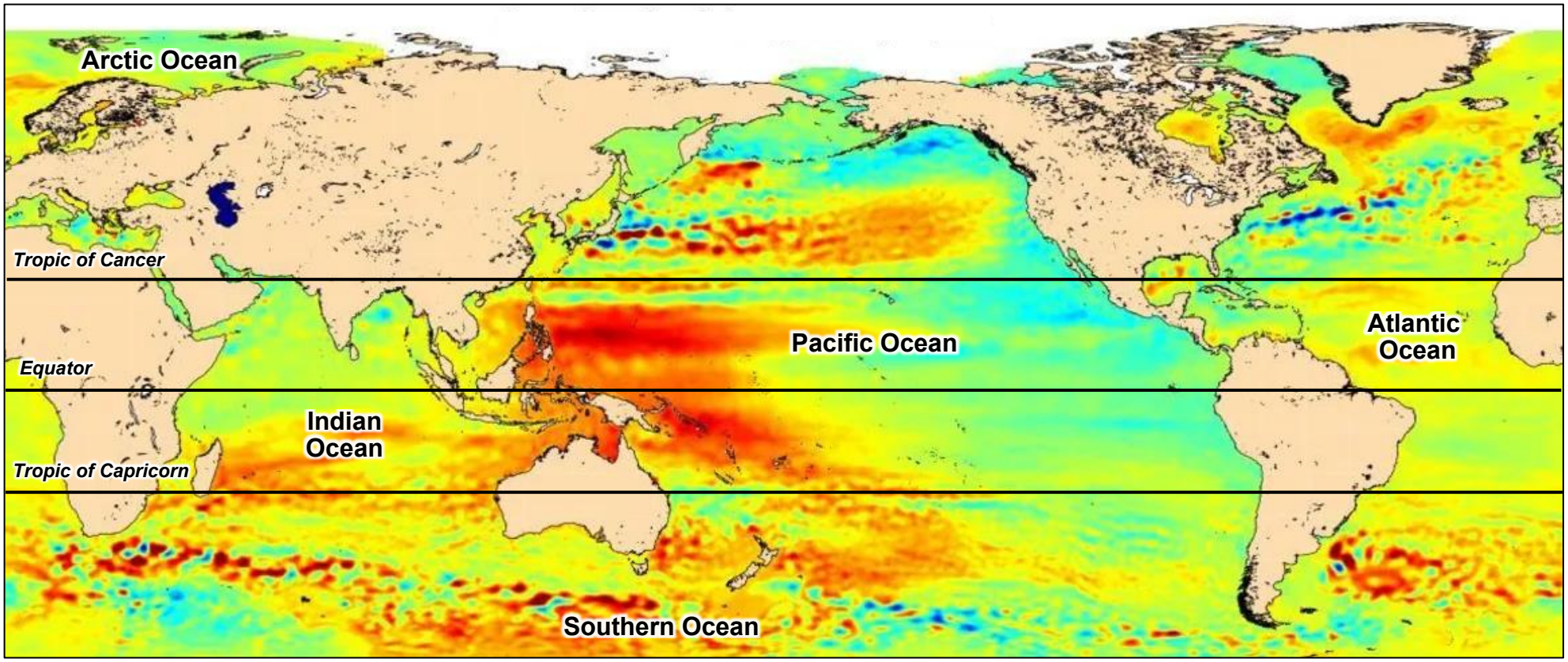
Resource 7 for Question 2

A 'highly deprived' slum in the Lagos metropolis, Nigeria



Resource 8 for Question 2

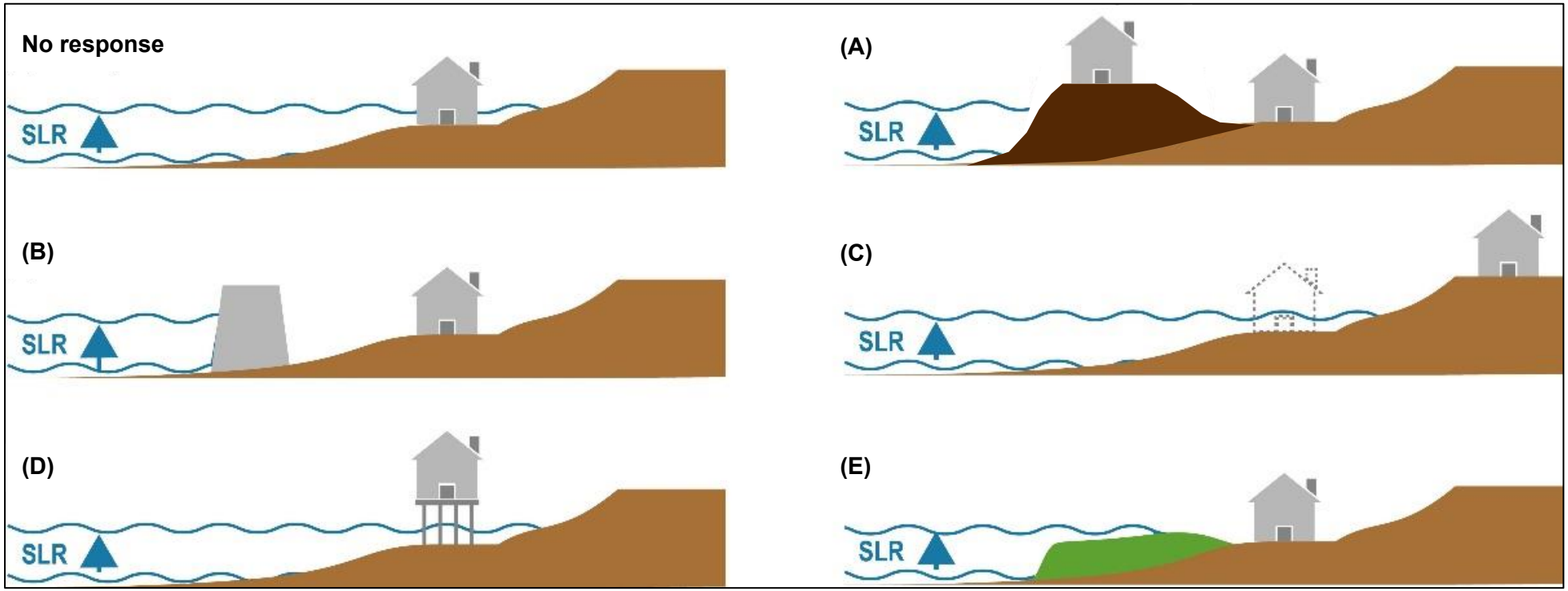
Mean annual global sea level change from 1992 to 2010



Mean annual global sea level change in mm

Resource 9 for Question 2

Strategies to manage sea level rise in the low-lying South Pacific islands



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Copyright Acknowledgements

Question 1	Resource 1	© Urban Redevelopment Authority. "Urban Redevelopment Authority Singapore website." Accessed 10 August 2025. https://eservice.ura.gov.sg/maps/.
Question 1	Resource 2	© Wzhkevin. "Sungei Bedok canal, taken from point it passes under Upper Changi Road East." Wikimedia Commons image. Accessed 10 August 2025. https://commons.wikimedia.org/wiki/File:Sungei_Bedok.jpg.
Question 2	Resource 5	© Lo, K., E. Miller, P. Dworatzek, N. Basnet, J. Silva, J. L. Van Berkum, R. B. Halldórsdóttir, and M. D. R. Dyck. *National Ecological Footprint and Biocapacity Accounts, 2025 Edition: Data and Metadata Version 1.0.* Footprint Data Foundation, 2025, https://footprint.info.yorku.ca/data/.
Question 2	Resource 6A, 6B and 6C	© Akoteyon, Isaiah Sewanu, Ibrahim Rotimi Aliu, and Olayemi Soladoye. "Household Levels of Deprivation to WaSH and Residential Conditions in Slum Settlements of Lagos, Nigeria." *Journal of Water, Sanitation and Hygiene for Development*, vol. 11, no. 1, 2021, pp. 60–74.
Question 2	Resource 7	© "Rangée de Bâtiments dans un Bidonville." *Pikbest*, https://fr.pikbest.com/backgrounds/rowing-row-of-buildings-in-a-slum_9581265.html. Accessed 26 Aug. 2025.
Question 2	Resource 8	© "Sea Level Rise." *CNN*, 2 Oct. 2012, https://edition.cnn.com/2012/10/02/tech/satellites-sea-level-rise/index.html. Accessed 26 Aug. 2025.
Question 2	Resource 9	© Oppenheimer, M., B. C. Glavovic, J. Hinkel, R. van de Wal, A. K. Magnan, A. Abd-Elgawad, R. Cai, M. Cifuentes-Jara, R. M. DeConto, T. Ghosh, J. Hay, F. Isla, B. Marzeion, B. Meyssignac, and Z. Sebesvari. "Sea Level Rise and Implications for Low-Lying Islands, Coasts and Communities." *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*, edited by H.-O. Pörtner, D. C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, and N. M. Weyer, Cambridge University Press, 2019, pp. 321–445. Cambridge, UK, and New York, NY. https://doi.org/10.1017/9781009157964.006.